KUREHA YAMAGUCHI

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PROFILE

I'm a Research Fellow in the current MATS cohort (systems security track), building red-blue teaming infrastructure to support continuous hardening of sandboxes. We're finding and exploiting novel sandbox-escape vulnerabilities in gVisor, Firecracker, Kata Containers etc using multi-agent configurations coupled with fuzzing and formal verification tooling. Before MATS, I was an AI Security Research Engineer for 3 years working on problems of priority to our UK Defence and National Security partners. Much of my work from this period is not publicly released but it included backdooring robotics, shipping security-critical software to stakeholders, and evading malware classifiers in mobile apps. Before this, I completed my undergrad and masters in Information and Computer Engineering from The University of Cambridge, supported by a 4-year Scholarship.

SELECTED RESEARCH & PROJECTS

PUBLICATIONS

Where Do Reasoning Models Refuse? ↗
Kureha Yamaguchi, Benjamin Etheridge, and Andy Arditi
ICML MechInterp Workshop · 2026

Addressing the Root Cause: Automated Fuzzing Harness Generator for Patch Completeness Testing ↗
Kureha Yamaguchi, Hanna Foerster, Jack Richings, Robert Mullins
Applied ML for Cyber Security Conference · 2026

2DSig-Detect: a semi-supervised framework for anomaly detection on image data using 2D-signatures ↗
Xinheng Xie, **Kureha Yamaguchi**, Margaux Leblanc, Simon Malzard, Varun Chhabra, Victoria Nockles, Yue Wu
Pattern Recognition Journal, Elsevier · 2025

An AI Red Team Playbook ↗
Anna Raney, Shiri Bendelac, Keith Manville, Mike Tan, **Kureha Yamaguchi**
Proc. SPIE, Assurance & Security for AI-enabled Systems · 2024

An AI Blue Team Playbook ↗
Mike Tan, **Kureha Yamaguchi**, Anna Raney, Victoria Nockles, Margaux Leblanc, Shiri Bendelac
Proc. SPIE, Assurance & Security for AI-enabled Systems · 2024

PROJECTS

[In progress MATS project] Sandbox red–blue teaming infrastructure ↗
Hardening infra for gVisor, Firecracker, Kata Containers and other sandbox architectures: novel sandbox-escape discovery and exploitation with multi-agent configurations and specialised tooling.
MATS project, systems security track · 2026

Security-critical software for stakeholders
Production deployment of decision-relevant software for detecting emerging National Security risks, shipped during my time at The Alan Turing Institute. Not publicly released.
Security-critical software for stakeholders · 2025

Automated fuzzing harness generation for OSS-CRS ↗
Harness generation pipeline for variant analysis and patch semantic correctness; being integrated into open-source framework, OSS-CRS. We develop fuzzing metrics for root-cause coverage, and show automated fuzzing harness generator can be used for patch completeness testing.
Open-source tooling for OSS-CRS · 2026

CANARY: early-warning honey-net for autonomous red-team agents ↗
Canary-style honey-net that gathers cybersecurity intelligence on kill-chain TTPs and behavioural fingerprints from autonomous red-team agents.
BlueDot x Apart def/acc hackathon · 2025

MITRE ATLAS contributions ↗
Open-source ATLAS matrix contributions from our systematic evaluation of critical vulnerabilities in production AI government systems; during research collaboration with The MITRE Corporation.
Collaboration with The MITRE Corporation · 2024

Women in AI Security Workshop ↗
Cross-organisational workshop bringing together experts across government, academia and industry.
Co-organised event · 2024, 2025, 2026

RELEVANT EXPERIENCE

MATS Research — Sep 2026 - Present
Research Fellow — *Berkeley, US*

· Research Fellow in the current MATS cohort (systems security track), building red-blue teaming infrastructure to support continuous hardening of sandboxes, motivated by loss-of-control catastrophic risks from advanced AI systems

- Finding and exploiting novel sandbox-escape vulnerabilities in gVisor, Firecracker, Kata Containers etc with multi-agent configurations, fuzzing and formal verification; participating in Google's vulnerability reward program, kvmCTF
- During this time, I'm also supervising an MPhil student at The University of Cambridge

The Alan Turing Institute Sep 2023 - Sep 2026
Research Engineer *London, UK*

- Honed expertise in ML and Security, as an AI Security Research Engineer with 3 years' experience within the Turing's Defence and National Security programme
- Brought clarity and ownership to ambiguous research problems under the research agreement between GCHQ and The Alan Turing Institute: e.g developed adversarial evasion attacks against ML-based Android malware classifiers and identified sources of experimental bias in their evaluation
- Combined cyber domain knowledge and strong engineering skills for offensive security work, e.g. designing automated fuzzing harness generation pipeline for variant analysis and patch semantic correctness
- Led weekly sessions for Software Engineering best practices, putting theory into practice by writing robust and efficient production code for safety-critical, real-world deployment
- ❀ **Won first place** at the UK MoD x Google Cloud LLM Hackathon- the judging panel consisted of Senior HM Government leaders and Google's technology leadership, including GCHQ's Chief Data Scientist, Head of DAIC, and Google Cloud's Vice President of Engineering

Supervised Program for Alignment Research (SPAR) Feb 2025 - May 2025
Research Fellow *Virtual*

- Led applied mechanistic interpretability research under the mentorship of Andy Arditi (Anthropic Safety Fellow 2025), performing countless experiments, and publishing the research at ICML 2025 R2-FM Workshop and ICML 2026 Mechanistic Interpretability Workshop
- Uncovered where safety decisions are made in reasoning LLMs, using techniques like counterfactual importance sampling and steering vectors (working with DeepSeek, Qwen and OpenAI models)
- ❀ **Won first place $2,000 prize** for my research fellowship presentation

Okinawa Institute of Science and Technology Graduate University Jul 2021 - Sep 2021
Research Engineering Intern *Okinawa, Japan*

- Delivered practical embedded hardware and software support for multi-purpose, multi-sensor autonomous systems at the Neural Computation Unit under Professor Kenji Doya
- Developed software (Python, Java, Git) to efficiently and robustly implement the model-based policy search learning framework, 'Probabilistic Inference for Particle-Based Policy Search' (Parmas et al., 2018), for an ambitious real-world robot balancing task
- Exercised good coding practices when writing, testing and deploying production code, such as Git version control, technical documentation, issue tracking and unit testing

Cambridge University Robotics × Google DeepMind Dec 2019 - Aug 2020
AI Student Researcher *Cambridge, UK*

- Advanced Reinforcement Learning research in collaboration with Google DeepMind and the Bio-Inspired Robotics Laboratory, which required exercising my problem-solving skills and ability to learn quickly
- Collaborated with research colleagues and technical experts from Google DeepMind to apply novel AI algorithms, engineering and embedded hardware techniques

EDUCATION

University of Cambridge Oct 2018 - Jun 2022
Information & Computer Engineering, ***B.A.*** *&* ***M.A.*** *&* ***MEng*** *with Honours*

- Bachelor's and Master's modules covering areas of Computational Statistics, Machine Learning, Deep Learning, Statistical Signal Analysis, Data Transmission, Information Theory and Coding
- ❀ **Awarded the Diamond Jubilee Scholarship**, a 4-year merit-based scholarship awarded by The Institution of Engineering and Technology
- ❀ **Awarded The Cambridge Blue and Hawks Award** for sporting excellence, University Women's Tennis 1st Team; Captaincy (2019/20); Presidency (2020/21); University Tennis Coordinator (2021/22)

SKILLS

Languages	Python (fluent); C/C++, Go, Rust, Bash, Java, JavaScript, MATLAB, Julia
Security	libFuzzer, AFL++, Jazzer, fuzz-introspector, Clang/LLVM sanitizers, tracee
ML/Infra	PyTorch, JAX, vLLM, CUDA, NNsight, TransformerLens, Inspect AI, GCP, Packer, Kubernetes, QEMU/KVM, vibe-coding workflow (alignment-hive, Herdr)
Natural languages	English (native), Japanese (fluent)